
NATIONAL UNIVERSITY OF SINGAPORE
IT5002 —Computer Systems and Applications
Final Assessment
Semester 1 AY2025/26

Time Allowed: 2 Hours

INSTRUCTIONS TO STUDENTS

1. Answer only on Exemplify. Do not write your answers in this assessment paper.
2. This assessment paper contains **THIRTEEN** questions and comprises **EIGHT** printed pages.
3. Students are required to answer **ALL** questions.
4. This is a **CLOSED BOOK** assessment.
5. You are allowed a single A4 cheat sheet printed/handwritten on both sides.
6. You are allowed a laptop to answer questions, and a stand-alone calculator. You are not allowed any other devices other than your laptop and calculator.
7. This paper is worth 50 MARKS.

Section 1. Multiple Choice Questions (20 MARKS)

There are 10 questions in this section, each worth two marks. Therefore this section totals 20 marks. Each question has exactly one correct answer. If you feel there is ambiguity, choose what you think is the best option.

1. Which ONE of the following statements regarding pipelining is FALSE?
 - a. The components within stages may produce their results at different times.
 - b. The time that the results of each stage is written to the pipeline registers depends on the system clock.
 - c. **+Pipelining shortens the execution time of individual instructions.**
 - d. Pipelining improves the throughput of instruction execution.
 - e. ALL of the above statements a. to d. are TRUE.

2. What is the main advantage of multicycle datapaths?
 - a. **+We can skip stages that are not used in an instruction.**
 - b. Every stage can operate at its own speed.
 - c. It is generally faster than single-cycle datapaths.
 - d. The number of cycles per instruction (CPI) is constant across all types of instructions.
 - e. None of the above options a. to d. are advantages of multicycle datapaths.

3. If we have an L1 cache with a hit rate of 95% and an access time of 3 ns, L2 cache with a hit rate of 99% and an access time of 10 ns and DRAM main memory with access time of 80 ns, what is the average memory access time?

$AMAT = 0.95 \times 3 + 0.05 \times (0.99 \times 10 + 0.01 \times (80 + 10)) = 3.39 \text{ ns}$

 - a. 2.17 ns
 - b. 3.25 ns
 - c. **+3.39 ns**
 - d. 4.15 ns
 - e. None of the above options a. to d. are correct.

4. Which ONE of the following statements is TRUE a memory cache with M cache blocks?
 - a. A direct-mapped cache is equivalent to an M-way set-associative cache.
 - b. A fully-associative cache is equivalent to a 1-way set-associative cache.
 - c. An n-way set associative caches reduce compulsory misses compared to a direct mapped cache. ($1 < n < M$)
 - d. **+An n-way set associative caches reduces conflict misses compared to a direct mapped cache. ($1 < n < M$)**

e. None of the options a. to d. are TRUE.

5. We have a 8 KiB direct-mapped data cache in a system with 64 KiB of memory (1 KiB = 1024 bytes) with 16-byte blocks. Words are 4 bytes long. Assume that we have this code:

```
for(i=0; i<100; i++)
    x = A[i];
```

Assuming that array A starts at address 0xAC00 and each array element is 4 bytes (i.e. 1 word), what is the hit rate of the data cache executing the program above?

Cache size = 2^{13}

Block size = 2^4

Total blocks = 2^9

Address layout: Offset = 4 bits, index = 9 bits, tag = $16 - 9 - 4 = 3$ bits

Array start = 101 01100 0000 0000

There are 4 words per block. We need 25 blocks. No conflict misses. First word will miss, subsequent 3 accesses will hit. Hit rate = 75%

- a. 28%
- b. 46%
- c. **+75%**
- d. 97%
- e. None of the above options a. to d. are correct.
6. Our system has a 32-bit addressing space, and a cache of 64 KiB (1 KiB = 1024 bytes), configured as a 4-way set-associative cache with 8 byte blocks. Words are 4 bytes long. Assume that each block holds, in addition to the 8 bytes of data, overhead in the form of the tag, a valid bit and a dirty bit. How large is the cache in total including the overhead? (Remember: Round up if your overhead per block does not work out to be an integer. E.g. if your overhead in a block is 3.25 bytes, round up to 4 bytes).

Each block = 8 bytes. Number of blocks = $\frac{2^{16}}{2^3} = 2^{13}$ blocks.

Number of sets = $\frac{2^{13}}{2^2} = 2^{11}$ sets.

Offset bits = 3, set index bits = 11, tag = $32 - 11 - 3 = 18$ bits.

Add in dirty and valid bits = 20 bits = $2.5 = 3$ bytes

Total overheads = $2^{13} \times 3 = 24576$ bytes

Add in 64 KiB = 90112 bytes

- a. 75 KiB

- b. +88 KiB**
- c. 92 KiB
- d. 1000 KiB
- e. None of the options a. to d. are correct.

7. Which of the following statements about kernels is TRUE?

- i. When making system calls to the OS, we need to switch to kernel mode to protect the user program from the OS.
 - ii. When making system calls to the OS, the need to switch to kernel mode makes such calls slow.
 - iii. "Process" is just another name for "program" and there is fundamentally nothing different between them.
 - iv. It is possible to add/remove components in monolithic kernels.
 - v. Microkernels are generally less reliable than monolithic kernels.
- a. +ii. and iv. ONLY are TRUE.**
 - b. i., ii. and v. ONLY are TRUE.
 - c. iii. and iv. ONLY are TRUE.
 - d. ii., iii. and iv. ONLY are TRUE.
 - e. None of the options a. to d. are correct.

8. Which ONE of the following show a valid series of state transitions for a process?

- a. New->Running->Blocked->Ready->Terminated.
- b. +New->Ready->Running->Blocked->Ready->Running->Terminated.**
- c. New->Ready->Running->Blocked->Running->Terminated.
- d. New->Running->Blocked->Ready->Running->Terminated.
- e. None of the above options a. to d. are valid.

9. We are given four processes P1, P2, P3 and P4 with the following characteristics, in a system with fixed priority scheduling.

In the execution profile, Cx-IOy-Cz means that the process spends x cycles on CPU, then y cycles in I/O and then z cycles again on CPU. Assume that there is only 1 I/O, that access to I/O is on first-come-first-served basis, and that there is no pre-emption on I/O.

So C4-IO12-C5 means that the process would spend 4 cycles on CPU, 12 cycles on IO then another 5 cycles on CPU.

Process	Starting time	Priority (0 = highest, 3 = lowest)	Execution Profile
P1	0	3	C3-IO2-C1
P2	2	1	C2-IO1-C2-IO3-C1
P3	1	2	C8
P4	0	0	C2-IO3-C3

Which process will be the last to finish?

- a. **+Process P1**
 - b. Process P2
 - c. Process P3
 - d. Process P4
 - e. Too little information to tell.
10. In a system with 1 MiB (1 MiB = 1024 KiB, 1 KiB = 1024 bytes) of memory with 1 KiB minimum allocation unit size with buddy allocation, what is the maximum number of steps that is needed to allocate a block for a request of 192 KiB?
- 1024 ->512/512
- 512->256

- a. 1 step
- b. **+2 steps**
- c. 3 steps
- d. 4 steps
- e. None of the above options a. to d. are correct.

Section 2. Long Questions (30 MARKS)

There are three questions in this section. Complete all parts of all questions.

11. (10 marks) We are given the following:

Program:

\$3: Base address for array A

\$4: Number of elements in A. \$4 contains the value 20.

```

                sll $4, $4, 2
                add $5, $3, $4
loop:          slt $6, $3, $5
                beq $6, $zero, exit
                lw $7, 0($3)
                addi $7, $7, 5
                sw $7, 0($3)
                addi $3, $3, 4
                j loop
exit:          .. Other irrelevant instructions ..

```

This table shows the timing of each stage:

Stage	Fetch	Decode	Execute	Memory	Writeback
Timing (ns)	2	1	2	3	1

a. (2 marks) How many instructions are executed by our program above?

Instructions before loop: 2

Instructions within loop: $7 \times 20 + 2 = 142$

Total = 144 instructions.

b. (2 marks) Assuming that your answer to part a. is 192 instructions, how much time would it take to execute this program on a single-cycle datapath? Express your answer in ns ($1 \text{ ns} = 10^{-9} \text{ s}$)

Total cycle time = $2 + 1 + 2 + 3 + 1 = 9 \text{ ns}$

Total time = $192 \times 9 = 1728 \text{ ns}$

The table below shows the stages that each class of instruction passes through (blank = stage is skipped)

Instruction Class	Fetch	Decode	Execute	Memory	Writeback
Arithmetic Instructions (including shifts)	X	X	X		X
Branch/Jump Instructions	X	X	X		
Load instructions	X	X	X	X	X
Store instructions	X	X	X	X	

As before 192 instructions are executed, broken down as follows by instruction class:

Instruction Class	Count
Arithmetic Instructions (including shifts)	85
Branch/Jump Instructions	45
Load instructions	31
Store instructions	31
TOTAL:	192

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Arithmetic Instructions (including shifts)	85
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Store instructions	31
TOTAL:	192

- c. (4 marks) What is the average number of cycles per instruction? Express your answer to 2 decimal places (e.g. 3.23, 4.15, etc)

$$\text{Average CPI} = \frac{85}{192} \times 4 + \frac{45}{192} \times 3 + \frac{31}{192} \times 5 + \frac{31}{192} \times 4$$

$$= 3.93$$

- d. (2 marks) If your answer to part c. is 3.75, how long does it take to execute the program above? Again assume that your answer to part a is 192 instructions. Express your answer in ns to 2 decimal places.

$$\text{Time per cycle} = \max(2, 1, 2, 3, 1) = 3 \text{ ns}$$

$$\text{Time} = 192 \times 3.75 \times 3 = 2160.00 \text{ ns}$$

12. (10 MARKS) We have the following program:

```

                                addi $3, $0, 0 # At address 0xCF0
                                addi $5, $0, 0
loop:                          andi $4, $3, 1
                                beq $4, $0, skip
                                addi $5, $5, 1
skip:                          addi $3, $3, 1
                                slti $4, $3, 10
                                bne $4, $0, loop

```

This program is run on a system with a 12-bit addressing space and an instruction cache with the following specifications

Cache Size	128 bytes
Block size	16 bytes
Type	2-way set associative

a. (3 marks) How many bits are in the tag portion of one block of this cache?

$$\# \text{ of blocks} = \frac{128}{16} = 8 \text{ blocks}$$

$$\# \text{ of sets} = \frac{8}{2} = 4 \text{ sets}$$

Set index = 2 bits

Offset = 4 bits

$$\text{Tag} = 12 - 2 - 4 = 6 \text{ bits.}$$

b. (4 marks) If we run the code above, what is the hit rate of this cache? Express your answer in percent to 2 decimal places (e.g. 10.42%). No marks will be credited if you do not express in this form even if your answer is correct.

$$\text{Address} = 0xCF0 = 110011 \ 11 \ 0000$$

		set-block
	addi \$3, \$0, 0 # At address 0xCF0	3-1
	addi \$5, \$0, 0	3-1
loop:	andi \$4, \$3, 1	3-1
	beq \$4, \$0, skip	3-1
	addi \$5, \$5, 1	4-1

```

skip:      addi $3, $3, 1           4-1
           slti $4, $3, 10        4-1
           bne $4, $0, loop       4-1

```

of memory accesses = 57

Cache misses = 2

Cache hits = 55

Hit rate = $\frac{55}{57} = 96.49\%$

- c. (3 marks) If your answer to part b. was 95%, and if a total of 70 instructions was executed at the end of this program, what is the TOTAL memory access time, assuming that the access time of the cache is 2ns while the access time for RAM is 80ns? Express your answer in nanoseconds.

$AMAT = 0.95 \times 2 + 0.05 \times 82$

$= 6 \text{ ns}$

Total memory access time = $5.9 \times 70 = 420 \text{ ns}$

13. (10 MARKS) A barber shop has 8 chairs each with a barber, and thus they can cut the hair of 8 persons at a time. Bob, the owner of the shop, has asked you to implement a program that will track the number of people coming in. If there are fewer than 8 persons, a new customer is able to enter immediately. If there are 8 people, then new customers are blocked from entering. When a customer leaves, another customer gets to enter.

You've decided to implement the system with semaphores. You have the following functions:

enter_shop	1. Blocks if 8 people are already in the shop. 2. Increment the number of people in the shop by 1. 3. Print the number of people now in the shop. 4. Return.
exit_shop	1. Decrement the number of people in the shop by 1. 2. Print the number of people in the shop. 3. Allow a waiting customer to enter, if any. 4. Exit.

We have the following operations:

```

sema_t sem=x;      // Create a counting semaphore called sem with initial value x
bin_sema_t bsem=x; // Create a binary semaphore called bsem with initial value x.
post(sem);         // Unblocks a process blocking on sem if any, otherwise increments sem.

```

```

// If sem is a binary semaphore, the maximum value it be increment
// to is 1.
pend(sem);    // Decrements sem if sem>0. Otherwise it blocks the calling process.

```

Complete the following code by choosing the most appropriate option.

```

int customer_count=0;

// Create mutex
bin_sema_t mutex=1;

// Create slots semaphore
sema_t slots=8;

void enter_shop() {
    pend(slots);
    pend(mutex);
    customer_count++;
    printf("ENTER: Number of customers: %d\n", customer_count);
    post(mutex);
}

void exit_shop() {
    pend(mutex);
    if(customer_count > 0)
        customer_count--;
    post(mutex);
    printf("EXIT: Number of customers: %d\n", customer_count);
    post(slots);
}

```

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Read y	P1	P3P1	P3P1	P3P1	P1	P3P1	P2P3P1	P2P3P1	P3P1	P3P1	P1	P1	P1	P3P1	P1	P1	P1	P1						
CPU	P4	P4	P2	P2	P3	P4	P4	P4	P2	P2	P3	P3	P3	P2	P3	P3	P3	P3	P1	P1	P1			P1
IO			P4	P4	P4	P2					P2	P2	P2									P1	P1	

P13	C3-IO2-C1	C3-IO2-C1	C3-IO2-C1	C3-IO2-C1	C3-IO2-C1	C3-IO2-C1	C3-IO2-C1	C3-IO2-C1	C3-IO2-C1	C3-IO2-C1	C3-IO2-C1	C3-IO2-C1	C3-IO2-C1	C3-IO2-C1	C3-IO2-C1	C3-IO2-C1	C3-IO2-C1	C3-IO2-C1	C2-IO2-C1	C1-IO2-C1	IO2-C1	IO1-C1	C1	-
P21	C2-IO1-C2-IO3-C1	C2-IO1-C2-IO3-C1	C1-IO1-C2-IO3-C1	IO1-C2-IO3-C1	IO1-C2-IO3-C1	C2-IO3-C1	C2-IO3-C1	C2-IO3-C1	C1-IO3-C1	IO3-C1	IO2-C1	IO1-C1	C1	-										
P32	C8	C8	C8	C8	C7	C7	C7	C7	C7	C7	C6	C5	C4	C4	C3	C2	C1	-						
P40	C1-IO3-C3	IO3-C3	IO2-C3	IO1-C3	C3	C2	C1	-																